2011 SRJC H2 Preliminary Examination Paper 2 Marker's Comments

1 Planning (P)

You are required to calculate the enthalpy change of neutralisation, $\Delta H_{neutralisation}$, for the reaction between solid citric acid, $C(OH)(COOH)(CH_2COOH)_2{\cdot}H_2O$ and aqueous sodium hydroxide.

Citric acid is a triprotic acid which can be represented as H_3A. The equation for the reaction is as follows:

$$H_3A + 3NaOH \rightarrow Na_3A + 3H_2O \qquad \textit{reaction 1}$$

FA 1 contains an aqueous solution of sodium hydroxide, NaOH, of unknown concentration. You may assume that the NaOH is in excess.

FA 2 contains 3.000 g of solid citric acid.

(a) Write a plan for the determination of the enthalpy change of neutralisation between **FA 1** and **FA 2**. You may use the reagents and apparatus normally found in a school or college laboratory.

In your plan you should give:

the essential details of the calorimetric procedure;

all appropriate tables to record the readings;

safety precaution(s) that are employed to minimise the spillage of the chemicals during the experiment.

[6]

1. Weigh about 3.000 g of citric acid / **FA 2** in a clean, dry weighing bottle and record in Table 1.
2. **Place the styrofoam cup in a 250 cm^3 beaker for support during the experiment.** [Safety precaution, see below]
3. Measure 50 cm^3 of **FA 1** in a measuring cylinder and pour it into the styrofoam cup.
4. Record the initial temperature of **FA 1**.
5. Pour **FA 2** into the solution in the styrofoam cup, stir the solution in the styrofoam cup to completely dissolve **FA 2** and record the highest temperature obtained.
6. Re-weigh the weighing bottle containing any residual solid **FA 2** and record in Table 1.
7. Record all temperature readings in Table 2.

mass of weighing bottle and **FA 2** /g	
mass of weighing bottle after pouring /g	
mass of **FA 2** (transferred) /g	

Table 1

Final temperature /°C	
Initial temperature /°C	
Temperature change /°C	

Table 2

(b) A student decides to modify the above experiment by using a solution of **FA 2** instead of solid **FA 2**, whilst keeping the total volume of solution mixture constant. Suggest how this modification would affect the experiment. Explain your answer. **[1]**

This will give the **true** temperature change between the neutralisation of citric acid and sodium hydroxide. **OR** It will make the experiment **more accurate**.
This is because the **enthalpy change of solution** of **FA 2** has **not** been factored in. Thus, the recorded temperature change will be different.

(c) Using the values that you have proposed in **(a)**, calculate the $\Delta H_{neutralisation}$ for *reaction 1* in terms of ΔT.
[4.3 J are required to raise the temperature of 1.0 cm^3 of solution by 1.0 °C] **[3]**

Heat change, $Q = 50 \times 4.3 \times \Delta T = 215\ \Delta T$ J

$\Delta H_{rxn} = -(215\ \Delta T) / (3/210.0) = -15050\ \Delta T$ J mol^{-1}

$\Delta H_{neutralisation} = -15050\ \Delta T / 3 = -5020\ \Delta T$ J mol^{-1}

Alternatively, $H_3A \equiv 3H_2O$
Amt of citric acid = 3/210.0 = 0.014286 mol
Amt of H_2O = 3 X 0.014286 = 0.042857 mol

$\Delta H = -\ 215\ \Delta T / 0.042857 = -\ 5020\ \Delta T$ J mol^{-1}

(d) To obtain an anhydrous sample of solid citric acid, a student proposes the following procedure:

Weigh the empty crucible provided. Then weigh it again with solid **FA 2** of about 0.100 g.
At first, gently warm the solid in the crucible otherwise the solid will "spit" out of the crucible. Then increase the heat gradually. Heat for about 8 minutes. Allow to cool on a heat proof mat, then reweigh. Then heat the crucible with the contents strongly for about 2 minutes. Repeat the heating, cooling, reweighing procedure until two consecutive weightings are within 0.010 g of each other.

(i) Suggest **one** potential problem that may arise should such a procedure be adopted and state how you would modify the procedure to minimise the problem.

The mass of solid used in 0.100g, which is **too little** for significant losses to be measured. The % error will thus be **significant.**
Increase the amount of solid used (e.g. 1.000 g)

(ii) It was noted by the student that the compound became black and charred after 8 minutes of heating. Suggest a reason why this might have occurred and state how you would modify the procedure to minimise the problem.

[2]

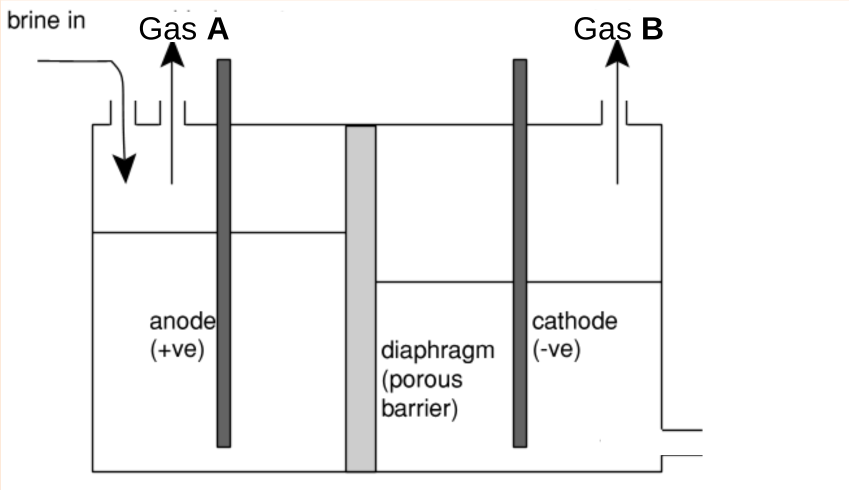
[Total: 12]

The citric acid is an organic compound. Thus, the compound can be <u>**oxidised easily / combustion / burnt**</u> at high temperature to result in the formation of carbon dioxide and water, with carbon deposits formed.
OR <u>**flame is too strong / heating temperature is too high**</u>

One way to minimise the problem is to conduct the heating in a <u>**water bath**</u> so that evaporation of the water can still occur.
OR Heat at a <u>**lower**</u> temperature / <u>**weaker**</u> flame
OR Heat for shorter intervals

2 This is a question on the halogens and their compounds.

(a) Chlorine can be manufactured by the electrolysis of brine, concentrated NaC*l*, using a diaphragm cell. During the electrolysis process, a steady current of 3.0 A was passed through the cell. 500 cm^3 of chlorine gas was collected at 30 ^oC and 1.5 atm.



(i) Write ionic equations for the reactions occurring at the anode and at the cathode.

Anode: $2Cl^- (aq) \rightarrow Cl_2 (g) + 2e$

Cathode : $2H_2O (l) + 2e \rightarrow 2OH^- (aq) + H_2 (g)$

(ii) Determine the amount of chlorine produced, stating any assumptions that you have made.

Assume Chlorine behaves **<u>ideally</u>**.

Using pV = *n*RT,

Amount of $Cl_2 = \dfrac{(1.5\times1.01\times10^5)(500\times10^{-6})}{(8.31)(273+30)}$ = 0.0301 mol

(iii) Hence, calculate the time taken for the process.

Since $2Cl^- (aq) \rightarrow Cl_2 (g) + 2e$

Amount of $Cl_2 = \dfrac{Q}{n_e F}$

$0.03008 = \dfrac{It}{n_e F}$

$0.03008 = \dfrac{3t}{2\times96500}$

$t = \dfrac{5805.44}{3}$ = **<u>1940 s</u>**

(iv) If the process was carried out at 5 °C, will your assumption in **(a)(ii)** still be valid? Why?

No.
This is because at low temperature, there is **significant intermolecular forces of attraction**. We can no longer assume that it is an ideal gas.

(v) With reference to relevant electrode potentials in the *Data Booklet*, explain why fluorine gas cannot be prepared by electrolysis of aqueous sodium fluoride, NaF(aq).

[11]

$F_2 + 2e \rightleftharpoons 2F^-$ +2.87 V
$O_2 + 4H^+ + 4e \rightleftharpoons 2\,H_2O$ +1.23 V
E^θ_{red} (F_2 / F^-) is much **more positive / bigger / higher** than E^θ_{red} (O_2 / H_2O)

O_2 will always be **preferentially discharged** (rather than F_2 gas).
Or F^- has **less tendency** to undergo oxidation compared to H_2O

(b) One major use of chlorine is in the bleaching process. When chlorine is reacted with aqueous sodium hydroxide at 20 °C, sodium chlorate(I), NaC*l*O is formed.

(i) Write a balanced equation for the reaction between chlorine and aqueous sodium hydroxide.

$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$
Or Cl_2 (aq) + 2NaOH (aq) $\rightarrow$ NaC*l* (aq) + NaC*l*O (aq) + H_2O (*l*)

(ii) In the reaction in **(b)(i)**, chlorine undergoes *disproportionation*. With reference to the changes in the oxidation states of chlorine, explain what is meant by the term in *italics*.

Cl_2 **increases its oxidation state from 0 to +1** (ClO^-) and **decrease its oxidation state from 0 to -1** (Cl^-) **at the same time/ simultaneously.**

(iii) Similar disproportionation occurs for bromine and iodine at even lower temperatures with the following observations:

At 0 °C,
Bromine reacts readily with aqueous NaOH to form BrO^- (+1).
Iodine reacts readily with aqueous NaOH to form IO_3^- (+5).

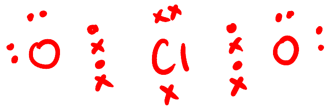
Suggest the relative stability of higher oxidation states from C*l* to I.

[3]

From C*l* to I: stability of high oxidation states **increases.** This is because the trend for electronegativity decrease from C*l* to I.

c) Manganese ions, Mn^{2+} (aq), present in water supplies can cause discolouration in laundered goods and deposits on industrial machines. Though chlorine can be used to control these problems, it reacts so slowly that Mn^{2+} (aq) ions may still be present in the water distribution system after 24 hours. Chlorine dioxide reacts much more rapidly in basic medium with Mn^{2+} (aq), oxidising it to manganese(IV) oxide. Chlorine dioxide is reduced to ClO_2^- (aq) ions.

(i) Draw the dot-and-cross diagram for chlorine dioxide and suggest a reason why chlorine dioxide is a good oxidising agent.



Chlorine dioxide has an **unpaired electron** thus it has a tendency to take in electrons.

(ii) With the aid of relevant half-equations, construct an equation for the overall reaction.

[4]

[R]: $ClO_2 + e \rightarrow ClO_2^-$

[O]: $4OH^- + Mn^{2+} \rightarrow MnO_2 + 2H_2O + 2e$

$2ClO_2 + 4OH^- + Mn^{2+} \rightarrow 2ClO_2^- + MnO_2 + 2H_2O$

(d) Account for the following observations:

(i) At around 30 °C, the RMM of hydrogen fluoride appears to be 40.0 and above 60 °C, it is about 20.0.

At low temperatures, **dimers** are formed as there are **strong/significant intermolecular hydrogen bonds.**

At higher temperature, the hydrogen bond breaks to yield single HF molecules.

(ii) Sodium chloride reacts with concentrated sulfuric acid to form hydrogen chloride while sodium bromide reacts with concentrated sulfuric acid to form bromine.

[2]

- Br^- is a **stronger reducing agent/ more easily oxidised** than Cl^-. Thus, conc. H_2SO_4 (aq) is strong enough an oxidising agent to oxidise HBr to produce Br_2. $2\,Br^- \rightarrow Br_2$
- It is too weak an oxidising agent to oxidise HC*l* to Cl_2

(e) The use of chlorine as a disinfectant in swimming pools is now widely banned and the weak acid trichloroisocyanuric acid is used instead.

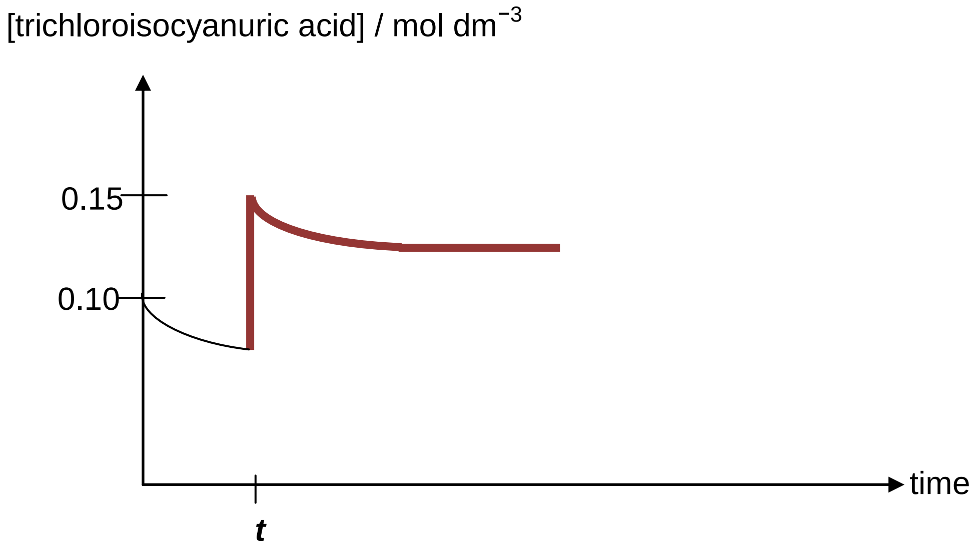
trichloroisocyanuric acid (aq) + OH^- (aq) $\rightleftharpoons$ ClO^- (aq) + cyanuric acid (aq)

The ClO^- (aq) is an effective disinfectant.

The graph below shows a decrease in the concentration of trichloroisocyanuric acid as the reaction progresses. At time ***t***, some trichloroisocyanuric acid is added until the concentration is increased to 0.15 mol dm^{-3}. Sketch, on the same axes, how the concentration of trichloroisocyanuric acid changes from time ***t***, till a new equilibrium is achieved.

[2]

[Total: 22]



3(a) A hydrocarbon burns in excess oxygen forming 480 cm^3 of carbon dioxide gas and 0.36 g of water at room temperature and pressure. It reacts with chlorine dissolved in tetrachloromethane to form a compound with only 1 chiral centre.

(i) Deduce the structure of a hydrocarbon containing four carbon atoms which fulfils the above requirements.

Amount of CO_2 = $\frac{480}{24000} = 0.02$ mol

Amount of C = 0.02 mol

Amount of H_2O = $\frac{0.36}{18.0} = 0.02$ mol

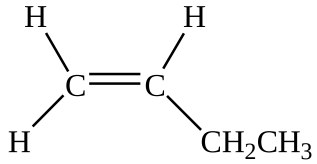
Amount of H = 0.04 mol

Mole ratio of C: H = 1:2

$\therefore$ Empirical formula of the hydrocarbon is CH_2.

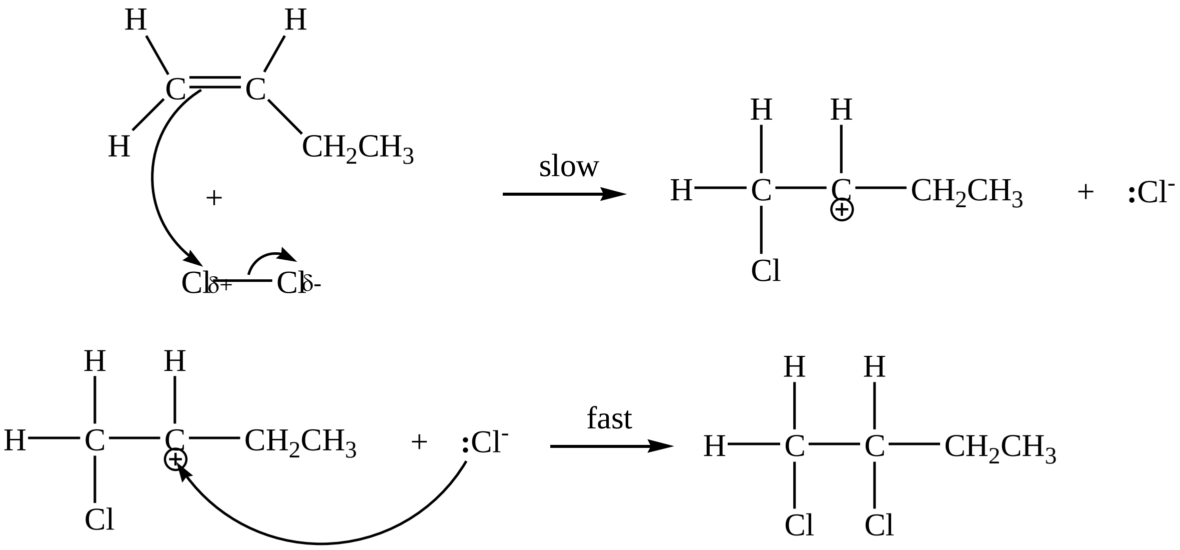
Since it has 4 carbon atoms, molecular formula = C_4H_8

Structure of hydrocarbon is



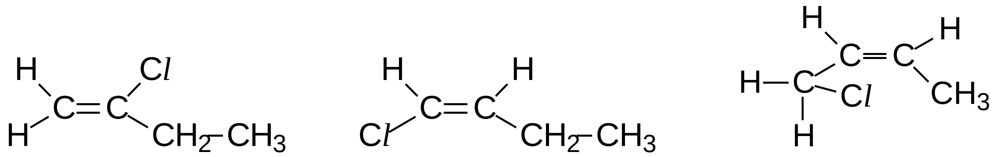
(ii) Name the type of reaction and describe the mechanism of the reaction between the hydrocarbon and chlorine.

Electrophilic addition.



(iii) The product from **(ii)** is then heated with ethanolic sodium hydroxide. Three different structural isomers of monochloroalkenes are obtained. Draw the structures of **any two** monochloroalkenes.

[7]

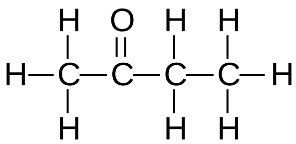


(b) Aldehydes and ketones are produced industrially by the catalytic oxidation of alkenes, e.g. ethanal is manufactured from ethene as shown below.

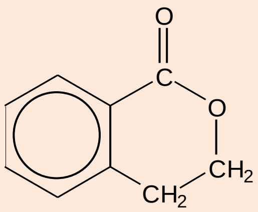
$CH_2{=}CH_2 + \frac{1}{2} O_2 \rightarrow CH_3CHO$

This process is also used industrially with but-2-ene. Draw the displayed formula of the product which would be produced from but-2-ene.

[1]



(c) Organic compounds can be distinguished using simple chemical tests. Suggest how you can distinguish between the following pair of isomers. State clearly the reagents and conditions used as well as the observations for each compound.



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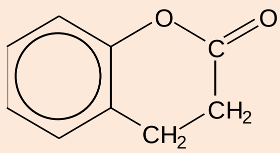


[2]
[Total: 10]

First, **add dil H_2SO_4 (aq)** to both compounds separately, and **heat.**
Then, cool and add **aqueous bromine / neutral aq. $FeCl_3$** to both samples.

For the 1st compound, the **reddish-brown bromine remains / no violet complex formed**, with the absence of the formation of any ppt.
For the 2nd compound, there would be a **decolourisation of reddish-brown bromine**, with the formation of a **white ppt / violet complex formed**.

Note: if use basic hydrolysis, must acidify the products before adding aq Br_2 or neutral $FeCl_3$ (aq)

Or

Add **$K_2Cr_2O_7$ (aq), dil H_2SO_4, heat**
(the acid will first hydrolyse the ester linkage, then the acidified $Cr_2O_7^{2-}$ will oxidize the primary alcohol that is produced from the first compound)
For the 1st compound, **orange** solution turns **green**.
For the 2nd compound, orange solution does **not** turn green.

4 The decomposition of azomethane, CH_3NNCH_3, is a homogenous and unimolecular reaction. When heated, azomethane decomposes to ethane and nitrogen as shown in the equation below:

$$CH_3NNCH_3\,(g) \rightarrow CH_3CH_3\,(g) + N_2\,(g) \qquad \Delta H_{rxn}$$

(a) Predict if the change in entropy of the reaction is positive or negative. Justify your answer. **[2]**

ΔS^θ is **positive** ; **Δn_{gas} = +1,** therefore more **disorder**.

(b) The following data are given:

Standard enthalpy change of formation of ethane (g) = -83.85 kJ mol^{-1}
Standard enthalpy change of formation of azomethane (g) = 148.8 kJ mol^{-1}

Use the given data to calculate the enthalpy change of reaction for the decomposition of azomethane.
[2]

$$\begin{array}{ccc} CH_3NNCH_3\,(g) & \xrightarrow{\Delta H_{rxn}} & CH_3CH_3\,(g) + N_2\,(g) \\ \Delta H_{f(azomethane)} \nwarrow & & \nearrow \Delta H_{f(ethane)} \\ & 2C\,(s) + 3H_2\,(g) + N_2\,(g) & \end{array}$$

$$\begin{aligned} \Delta H_{rxn} &= \Delta H_{f(ethane)} - \Delta H_{f(azomethane)} \\ &= -83.85 - 148.8 \\ &= -232.65 = -\mathbf{233\ kJ\ mol^{-1}} \end{aligned}$$

(c) A structural isomer of azomethane is $NH_2CH{=}CHNH_2$, identify the type of hybridisation about the C atom and explain why the π bond in $NH_2CH{=}CHNH_2$ is described as a localised π bond. **[2]**

sp^2
It is considered as a localised π bond as the **p electrons** involved are **restricted** between the **2 C atoms**.

(d) Explain, in terms of structure and bonding, why CH_3CH_3 is more soluble in tetrachloromethane than $NH_2CH{=}CHNH_2$. **[2]**
[Total: 8]

Both CH_3CH_3 and $NH_2CH{=}CHNH_2$ have **simple molecular structure**, with **intermolecular van der Waals' forces of attraction** and **intermolecular hydrogen bonds** respectively.

CH_3CH_3 can form **<u>favourable van der Waals' forces of attraction</u>** between solute and solvent molecules.

OR

<u>Weaker van der Waal's forces</u> between CCl_4 molecules are **<u>not strong</u>** enough to **<u>displace the stronger hydrogen bonding</u>** between $NH_2CH{=}CHNH_2$ molecules.

5 Elements **E**, **F** and **G** are period 3 elements.

(a) **E** is a white solid with melting point of 317 K. When exposed to air, it reacts spontaneously, producing a white powder of empirical formula $\mathbf{E}_2O_5$.

E$_2O_5$ readily undergoes neutralisation with aqueous NaOH to form salt and water.

(i) From the observations above, deduce the identity of **E**.

Low melting point hence E must be of simple molecular structure: sulfur or phosphorus

The oxide of E is a white powder with empirical formula E_2O_5 hence must be **phosphorus** as sulfur is yellow and does not have empirical formula of E_2O_5

(ii) Based on your deduced identity of **E**, describe the above reactions using only equations with state symbols.

P_4 (s) + 5 O_2 (g)→ P_4O_{10} (s)

P_4O_{10} (s) +12 NaOH(aq)→ 4 Na_3PO_4(aq) + 6 H_2O(*l*)

(iii) With the aid of an equation, suggest and account for a pH value for the resultant solution when the oxide of **E** dissolves in water.

[6]

P_4O_{10}(s) + 6H_2O(*l*)→ 4 H_3PO_4(aq)

Resulting pH: **2–3**

(b) Element **F** forms an **amphoteric** oxide readily in air. Using only equations with state symbols, explain what is meant by the word in bold.

[2]

Al_2O_3(s) + 6 HC*l*(aq) → 2 $AlCl_3$(aq) + 3 H_2O (*l*)

Al_2O_3(s) + 2 NaOH(aq) + 3 H_2O(*l*) → 2 Na[$Al(OH)_4$] (aq)

(c) The successive ionisation energies of element **G** is shown below.

	1st	2nd	3rd	4th	5th	6th	7^{th}	8^{th}
I.E. / $kJmol^{-1}$	999	2265	3331	5071	7008	8487	27107	31719

(i) Deduce the group number in which **G** belongs.

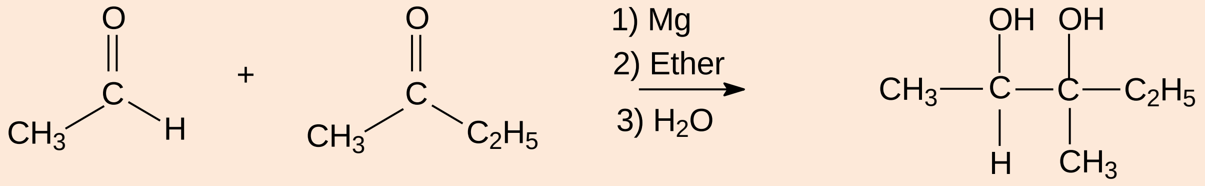
Group VI

(ii) Write down the electronic configuration of **G** in the +4 oxidation state.

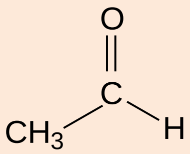
[2]
[Total: 10]

$1s^2 2s^2 2p^6 3s^2$

6 Two carbonyl groups can be coupled together to form a diol, as seen from the reaction below.



(a)(i) Draw the displayed formula of the diol that will be formed when the following compounds are reacted.



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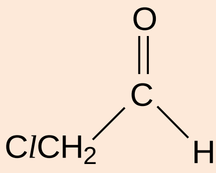
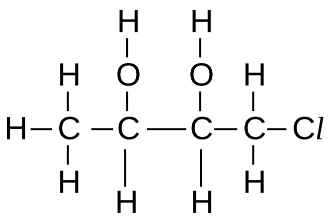
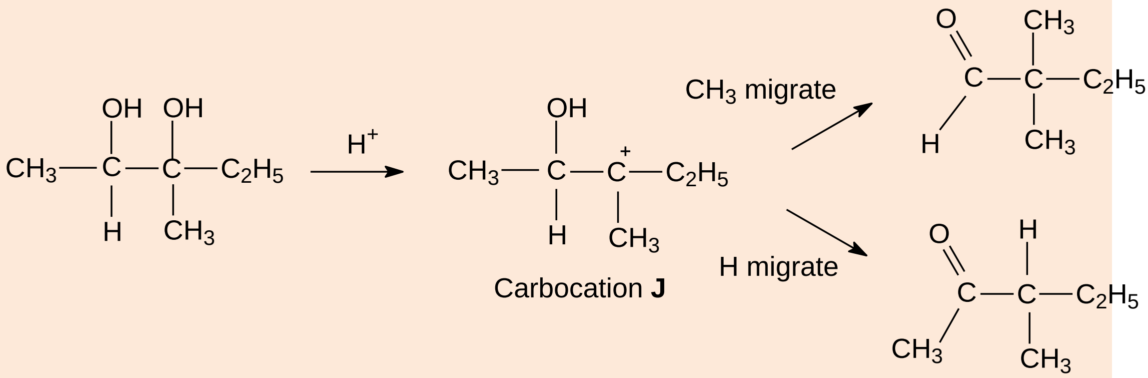
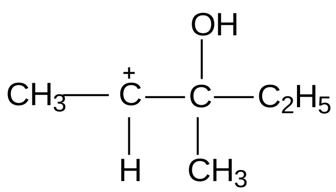




The Pinacol rearrangement involves the formation of carbonyl compounds from a diol via the following reaction scheme.



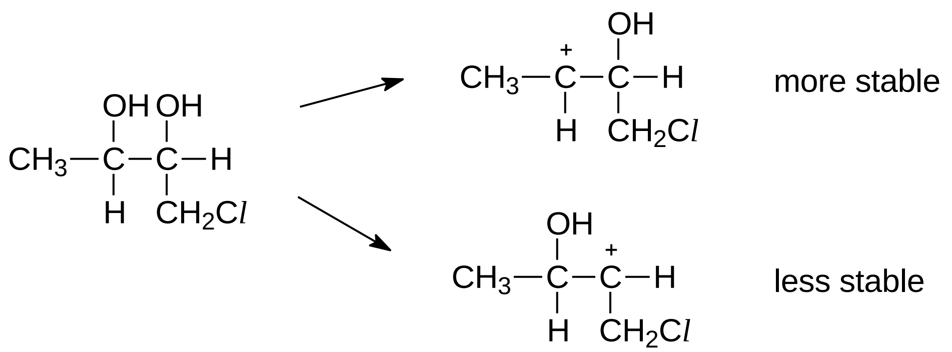
(ii) Another carbocation **K**, other than **J**, can also be formed. Draw the structural formula of **K** and suggest a reason why the yield of **K** is smaller.

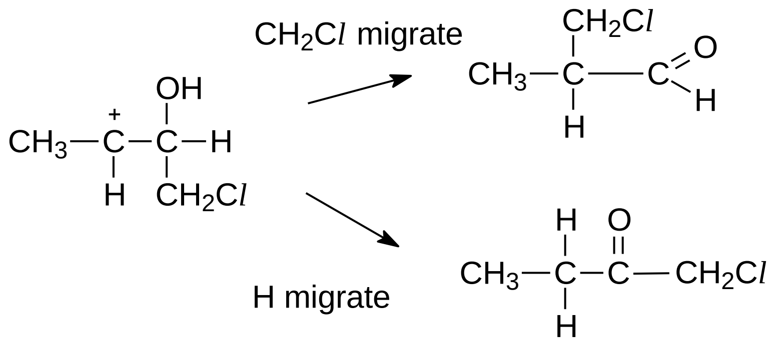


The yield of carbocation K is smaller as it is a **<u>less stable</u> carbocation**.

(iii) Draw the more stable carbonyl compounds formed when your diol in **(a)(i)** undergoes the Pinacol rearrangement.

[5]



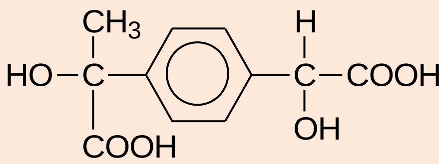


(b) A compound **L**, $C_9H_{10}Br_2$, undergoes reaction with aqueous sodium hydroxide under reflux to give compound **M**, $C_9H_{12}O_2$. Compound **M** then undergoes nucleophilic substitution with $ClCOCH_2COCl$ to form a cyclic compound **N**. Compound **M** gives a yellow precipitate when it undergoes oxidation with warm alkaline aqueous iodine.

Compound **M** can also undergo a series of reactions as shown below to yield Compound **Q**.

Compound **M** ⟶ Compound **O** ⟶ Compound **P**

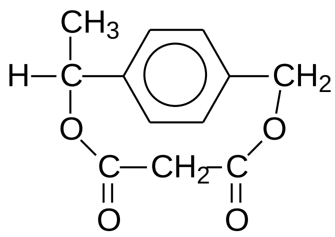
↓



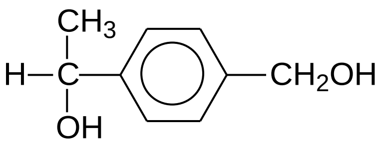
Compound **Q**

(i) Draw the structural formulae of compounds **M**, **N**, **O** and **P**.

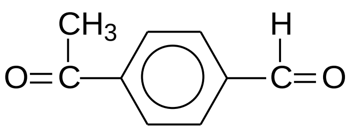
Compound N



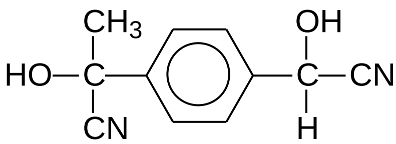
Compound M



Compound O



Compound P



(ii) State the reagents and conditions for converting **O** to **P**.

[5]

Reagents and conditions for step 4:

<u>HCN with trace amt of KCN/aqNaOH, cold (10 – 20^{o}C)</u>